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BioSMB Technology as an Enabler for a Fully Continuous Disposable Biomanufacturing Platform

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2.1

Introduction

Many process industries have implemented continuous processing during the twentieth century. One of the best known examples is the manufacturing of cars. Henry Ford installed a belt-driven moveable production line to enable continuous flow production of the famous Model T Ford. The manufacturing process of the car was broken down into 84 areas. The conveyor belt took the intermediate product from station to station where the assembly employees would perform their tasks. By implementing this system, the time to construct a car was reduced from 13 h to 93 min, while using less manpower. The costs went down from USD 850 to below USD 300. Mass production had made cars affordable for common people.

Other notable examples of continuous processing can be found in petrochemicals, pulp and paper, metal, and food industries. In all these areas, continuous manufacturing has enabled mass production, thereby realizing significant cost savings and making the products available to a larger population.

In most process industries, continuous production has also resulted in improved process control and improved safety. Reports indicate that 60–70% of the accidents in chemical industries occur during nonroutine operations such as startup and shut-down. As a result, continuous processes are inherently safer than batch processes, where startup and shut-down operations are required more often. The pharmaceutical industry has already started to gradually implement continuous manufacturing. For some large-scale products, such as over-the-counter (OTC) products, this has been a reality for a few decades now. In most cases, the choice for continuous processing was driven by the economies of scale. This has, for instance, also been driving many antibiotics manufacturers toward (semi-) continuous processes.

The manufacturing process of biopharmaceutical products involves multiples steps, all of which are conducted in batch mode. Although the biopharmaceutical industry utilizes state-of-the-art process chemistries to manufacture their drugs, the technological implementation of these processes lags behind compared to other mature manufacturing industries.

Until recently, the drivers for change were very limited due to a combination of reasons, including lack of competition, relatively small production volumes, and limitations in process understanding and process control. Some of these factors are now rapidly changing. Biosimilars are already a reality in Europe and it is inevitable that the impact of their presence will grow in the next decade. In addition to this, the demand for biopharmaceutical products and monoclonal antibodies in particular, is growing steadily and some products have exceeded ton-scale demands. Meanwhile, technological advancements have improved process understanding and the ability to measure and monitor the manufacturing processes has improved significantly. This transformation coincides with the US Food and Drug Administration (FDA) initiatives on quality by design (QbD) and process analytical technologies (PAT). With these trends in the market, the biopharmaceutical industry has started exploring continuous production strategies for recombinant proteins and monoclonal antibodies.

In this chapter, we will review which of the individual building blocks are needed to assemble an integrated continuous bioprocessing platform. In addition to this, the opportunities to combine continuous processing and disposable bioprocessing solutions will be addressed. The main focus of the chapter is on the downstream processing section of the production process. More in particular, the impact of the recent advancements in multicolumn chromatography processes will be discussed.

2.2

Integrated Continuous Bioprocessing

Today's manufacturing of biopharmaceutical products involves a cascade of batch processes. The cascade can be divided into two sections, being the upstream process (USP) and the downstream process (DSP). During the cascade of batch unit operations, the intermediate product solution is collected after each step. Only after completion of the process step the intermediate product solution is connected to the next process step. This batch-wise manufacturing procedure results in relatively low capacity utilization.

The potential benefits of an integrated continuous biomanufacturing approach include time savings and improved capital utilization. In an industry where the costs of goods are dominated by fixed costs, this represents a significant business driver. A typical scheme for a cascade of batch process steps is displayed in Figure 2.1a and for an integrated continuous biomanufacturing process is displayed in Figure 2.1b.

The schedules in Figure 2.1 show the following features of a continuous process:

- a) The overall processing time is determined by the slowest step. All other unit operations are sized to match this flow rate. In most downstream

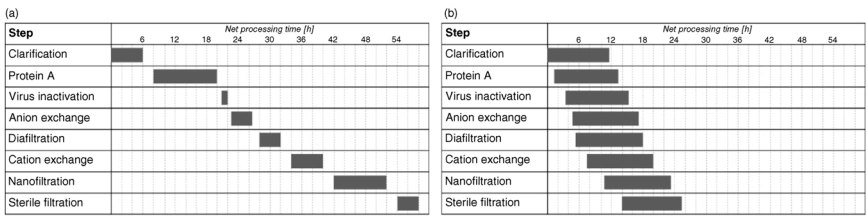


Figure 2.1 Typical DSP manufacturing schedules for (a) a cascade of batch-wise operated unit operations, and (b) an integrated continuous downstream process.

- processing cascades, the slowest process step is the capture chromatography step.
- b) As a result of the continuous flow from one unit operation into the next, the interstage product hold bags can be eliminated or minimized.
 - c) Each unit operation is only started as soon as the first product comes off the previous unit operation. As a result, there will be a cascaded startup and shut-down procedure.

In order to implement an integrated continuous biomanufacturing process, all unit operations need to be compatible with continuous flow of the product solution. The compatibility chart with this requirement for unit operations that are normally found in the manufacturing of glycosylated proteins or monoclonal antibodies is indicated in Table 2.1.

Based on existing technologies, one can identify a few unit operations that are not yet fully compatible with an integrated continuous bioprocessing cascade.

Virus inactivation relies on a residence time under certain conditions, most commonly reduced pH in a hold tank. Translating such process into a continuous process may be done in various ways. Brower had presented a strategy based on a continuous flow plug flow reactor [1]. Alternatives that rely on hold times in smaller sublots have been described elsewhere [2]. Thus far, none of these approaches have been validated, but there seems to be no reason to assume that virus inactivation cannot be transformed into a continuous process. The critical process parameters are the pH and the hold time and these can be adequately controlled in a continuous flow process too. Alternative flow-through virus inactivation procedures, such as the UV inactivation, were reported to be less effective for enveloped viruses, which is the main target for the low pH inactivation [3,4].

Table 2.1 Summary of typical DSP unit operations and their compatibility with continuous flow processing.

Step	Compatible with continuous processing	Examples and/or comments
Clarification	Yes	Continuous centrifugation Depth filtration
Capture chromatography	No	Protein A chromatography
Virus inactivation	Yes/no	Requires a cascade of mixers or a tubular contactor
Ultrafiltration	Yes	Single-pass Tangential flow filtration
Trace impurity removal	Yes	Q membrane adsorbers
Aggregate removal	No	Cation exchange chromatography Hydrophobic interreaction chromatography
Virus filtration	Yes	

The concept of multistage, single-pass tangential flow filtration has been used in other process industries, such as the dairy industry. A similar approach has recently been implemented in the CadenceTM single-pass tangential-flow filtration (TFF) technology [5]. Although multistage tangential flow filtration has also been used for diafiltration processes in other industries, the CadenceTM system seems to be mainly tested for continuous concentration processes.

Chromatographic purifications have been the workhorse for biopharmaceutical purification. A typical production process comprises three chromatography steps. Traditionally, chromatographic separations are performed in a single packed column. In traditional process industries, batch chromatographic separations are transformed into continuous processes by adopting the concept of multicolumn chromatography. This is the basis for simulated moving bed technology and all other continuous ion exchange and continuous chromatography processes that have been implemented in a wide range of process industries.

2.3

Multicolumn Chromatography

The principle of multicolumn chromatography is to create a (simulated) movement of the chromatography columns in opposite direction of the process solutions. This results in a countercurrent contact between the liquid and the chromatographic media, which allows overloading the columns beyond their dynamic binding capacity without suffering loss of material. When product breaks through from the first column, it will be captured on a second column in the load zone. With this, countercurrent chromatography processes can offer a significant gain in capacity utilization.

Another benefit of the countercurrent contact approach is that it eliminates idle zones in the process. In a batch chromatography column, the mass transfer zone only covers a small portion of the overall chromatographic media volume. The media above the mass transfer zone is in equilibrium with the feed solution and has no additional capacity to bind more product. The media below the mass transfer zone is in contact with depleted feed solution and hence is waiting for the first product to arrive. In a countercurrent process, these idle zones are eliminated and the load zone can be designed to only cover the length of the mass transfer zone. This generally corresponds to a small part of the batch column volume. These two features are schematically demonstrated in Figure 2.2.

In batch processes, the column size is proportional to the total mass of protein that needs to be purified and hence there is a direct relationship with the feed concentration. In a continuous chromatography process, the load zone is mainly designed around the contact time associated with mass transfer zone. The total volume of chromatographic media in the load zone thus hardly depends on the static binding capacity and the feed concentration. Instead, the process is designed around the volume that needs to be processed, or more precisely, the feed flow rate.

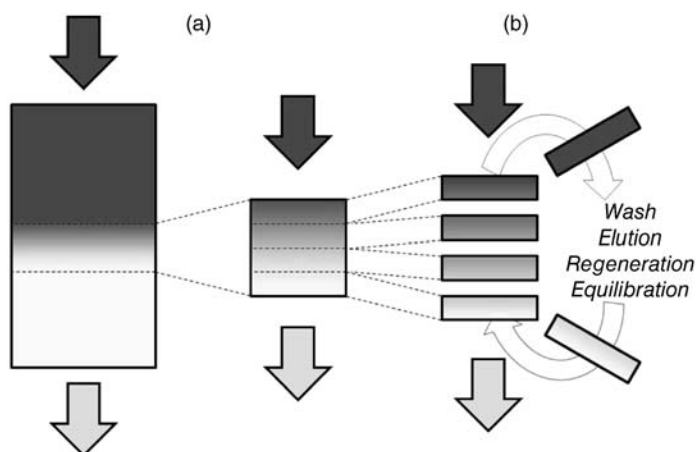


Figure 2.2 Schematic comparison between (a) a batch process and (b) a multicolumn countercurrent chromatography process.

2.4

BioSMB Technology

In order to provide a practical way to implement multicolumn chromatography in the biopharmaceutical industries, Tarpon Biosystems has developed the BioSMB[®] Technology that relies on a fully disposable flow path. All valves that are required to direct flows to and from all columns are kept within a single disposable cassette that can be mounted to the BioSMB system. The overall layout of the system is schematically shown in Figure 2.3.

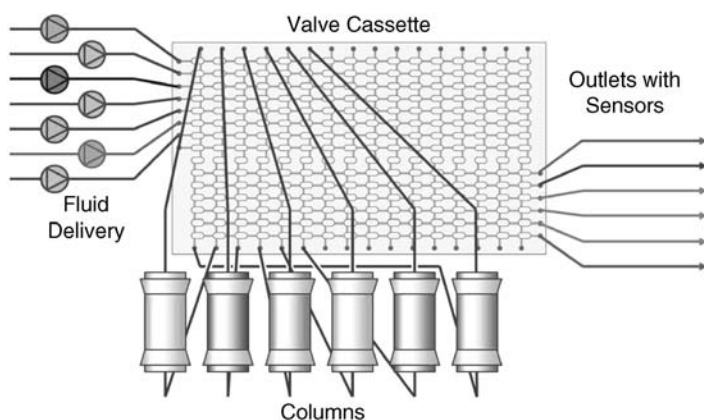


Figure 2.3 Schematic lay-out of the BioSMB[®] system for continuous multicolumn chromatographic separations.

Table 2.2 Some examples for which BioSMB technology has been successfully tested [6–11].

Case study	Chromatographic mode	Specific productivity
Capture of antibodies	Protein A chromatography	2–6 × batch
Aggregate removal	Hydrophobic interaction (HIC)	2–3 × batch
Aggregate removal	Ion exchange	4–8 × batch
Capture of recombinant proteins	Ion exchange	2–5 × batch
Capture of VLP vaccines	Ion exchange	3–7 × batch
Polishing of a VLP vaccines	Size Exclusion (SEC)	6–14 × batch

When the BioSMB system is combined with prepacked columns, membrane adsorbers or any other chromatographic devices designed for single-use applications, the entire chromatography process can be translated into a viable single-use option. With this, the BioSMB® Technology provides a promising answer for those companies who are developing completely disposable platforms for the entire biomanufacturing process.

For various chromatographic separations, proof of concept has been demonstrated. The majority of applications that have been investigated involve Protein A affinity chromatography for the purification of monoclonal antibodies [6–8]. Other notable examples include aggregate removal using cation exchange [9] or hydrophobic interaction chromatography [10] and vaccine purification using ion exchange and size exclusion chromatography [11]. In all these cases, the specific productivity gain was substantial (Table 2.2).

Continuous multicolumn chromatography processes can be designed to either operate at maximum specific productivity or at maximum capacity utilization. In order to operate at maximum specific productivity, the process should be designed for a minimum residence time in the load zone. This can be achieved by compromising on capacity utilization. This may well represent the most attractive design for clinical manufacturing, where the costs of the chromatographic media are dominated by the actual volume that needs to be installed. During a typical clinical manufacturing campaign, the resin is depreciated well before it has reached its maximum lifetime in terms of process cycles. The costs for chromatography media in clinical manufacturing are therefore governed by the amount of media installed. The lower capacity utilization is compensated by running more cycles per batch on each column.

During commercial or routine manufacturing of biopharmaceuticals, on the other hand, the chromatographic media is generally utilized to its full lifetime. This may involve 100 or 200 cycles, depending on the process and the validation constraints. As a result, the costs for chromatographic media per unit of purified product depends on the amount of product bound (and eluted) in each cycle per liter of chromatographic resin, which corresponds to the capacity utilization. Continuous chromatography for routine manufacturing should probably be optimized around maximum capacity utilization, which requires longer residence times than in the scenario listed earlier. This translates into larger column sizes

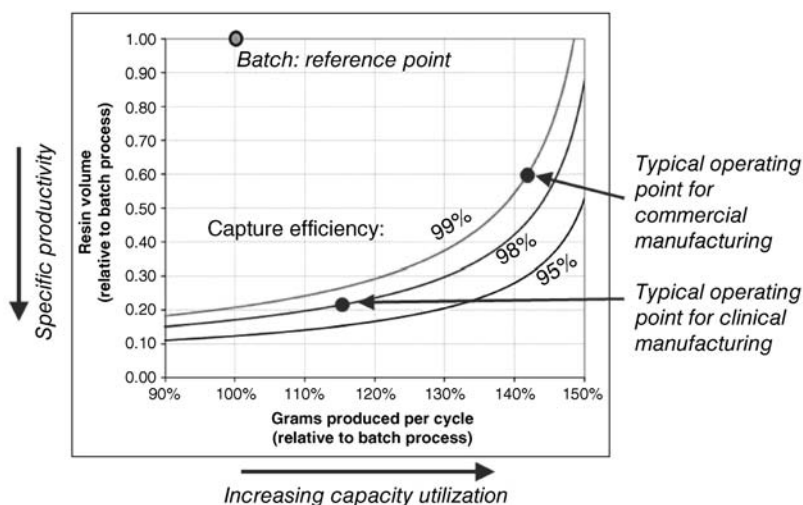


Figure 2.4 Schematic representation of the typical operating points for clinical manufacturing and commercial manufacturing in a Protein A based BioSMB process.

and hence the replacement frequency will be lower, yet still higher than with traditional batch chromatography.

Two typical operating points representing commercial and clinical manufacturing are shown in Figure 2.4. On the vertical axis, the relative volume compared to a typical batch process is shown. The design calculations underlying this graph have been presented elsewhere [7]. On the horizontal axis, the capacity utilization is plotted relative to the reference batch process. On the vertical axis, the amount of chromatographic media that is required for the process is plotted (relative to the reference batch process). The graph shows how one can decide to either operate at more or less the same capacity utilization as the batch process, with only 15–20% of the chromatographic media.

On the other hand, the preferred scenario for commercial manufacturing may lead to approximately 40–50% more protein produced in each cycle per liter of resin, thereby representing 35% cost savings. Even at that operating point, the process still requires significantly less chromatographic media than the batch process. In addition to this, the (smaller) amount of media is distributed over multiple columns, the size of which is significantly smaller than the batch equivalent. This allows a manufacturer to purchase prepacked columns, thus minimizing column packing operations and the associated costs.

A key aspect of continuous chromatography is that the amount of media installed is predominantly governed by the residence time that is required to reach the desired capacity utilization. This is mainly related to the mass transfer resistance (or diffusion) associated with the mass transfer process. Neither the titer nor the static binding capacity has a (huge) impact on the residence time.

The column size therefore mainly depends on the flow rate and therefore on the volume to be processed and the processing time. This gives operational flexibility in handling variations in titer in a facility. In various proof of concept studies, the ability of the BioSMB® technology to handle different feed solutions with the same equipment and column dimensions was already demonstrated [8,9]. In these studies, the antibody concentration in the feed stock was varied from approximately 2 to 10 mg ml⁻¹.

The number of columns that is required to run a continuous process does depend on titer. In order to transform the load step into a continuous counter-current step, at least two columns are needed. This brings the minimum number of columns for a continuous process to three, provided that one column provides sufficient time to do all wash steps, elution, regeneration and re-equilibration steps. As soon as the load volume becomes relatively low compared to total process time, which is the case for medium and higher titers, the load time becomes proportionally smaller and one column is no longer sufficient. This is schematically displayed in the circular chronograms in Figure 2.5. These chronograms display all steps in the chromatography process, where the inlet for each step is displayed on the outer ring and the outlet of each step is displayed on the inner ring. The columns are represented by the spokes that will travel clockwise over the chronogram. At increasing titer, the volume that can be loaded on the columns will be less and hence the load time will become shorter. Since the mass transfer kinetics is not (or hardly) affected by the increasing titer, the residence time in the load step will remain the same and hence the column volume can remain the same. The consequence of an increasing titer thus is that each column will spend less time in the load zone of the process cycle. The time

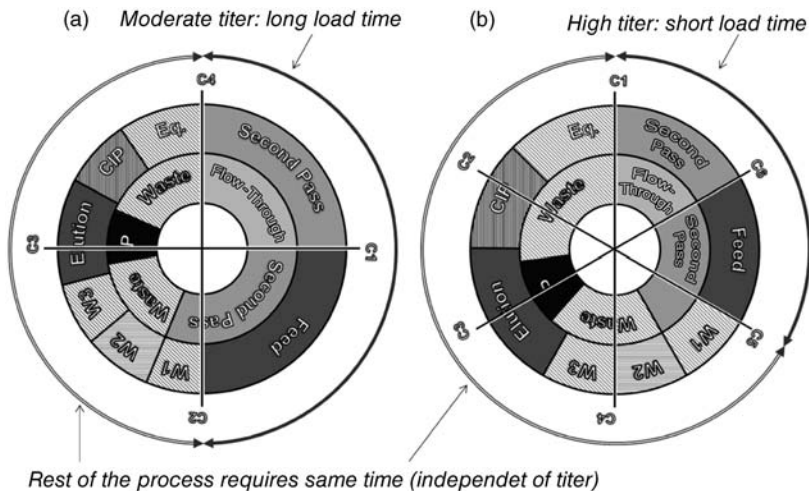


Figure 2.5 Circular chronogram for (a) a low-titer process with minimum number of columns and for (b) a high-titer process with a minimum number of columns.

required for the other steps in the process remains the same. Relative to the load time, however, the rest of the cycle therefore becomes longer. This needs to be compensated by adding extra columns to provide the required time in the rest of the cycle.

One of the key features of BioSMB® Technology is the ability to connect extra columns without adding complexity to the system. This is particularly beneficial in accommodating processes with different titers as discussed earlier. The flexibility in terms of the number of columns may also provide a means to enhance specific productivity by using more, yet smaller columns. This allows one to design a capture process with the same yield but with an overall lower inventory of resin, simply because the media is divided into smaller increments. Additional columns may also allow further optimization of the process in terms of advanced elution schemes to boost product titers without compromising on product yield. Last but not least, in general more complex separations may require more columns.

2.5

Fully Disposable Continuous Processing

Disposable and single-use bioprocessing technologies have been implemented throughout the entire field of biopharmaceutical manufacturing. Most notable examples are buffer containers, product containers, and single-use bioreactors. For downstream processing, filters and membrane adsorbers are most likely the best known examples of disposable technologies.

Table 2.3 lists the most common unit operations for the downstream processing of glycosylated proteins and monoclonal antibodies and their availability in a single-use or disposable format. The agreement between the compatibility tables

Table 2.3 Summary of typical DSP Unit Operations and their availability in a disposable or single-use format.

Step	Available in single-use or disposable format	Examples and/or comments
Clarification	Yes	Disposable centrifugation, depth filtration
Capture chromatography	No	Protein A chromatography
Virus inactivation	Yes	Regular hold bag
Ultrafiltration	Yes	Single-use TFF
Trace impurity removal	Yes	Q membrane adsorbers
Aggregate removal	No	Cation exchange chromatography
		Hydrophobic interreaction chromatography
Virus filtration	Yes	

shown in Tables 2.1 and 2.3 is striking. The nature of continuous flow unit operations is that the system can be smaller and hence disposable design becomes a viable option. For this reason, continuous chromatography is an enabler for designing a fully disposable downstream processing cascade.

Single-use or disposable multicolumn chromatography fills the main gap in the entire cascade and with that, BioSMB[®] Technology is an enabler for a fully continuous and fully disposable downstream process.

Such an integrated, fully continuous downstream processing scheme has recently been presented by Merck [12]. The upstream process was based on fed-batch cultivation. The downstream process utilizes two BioSMB[®] systems: the first system for Protein A chromatography and the second system for cation exchange. Processing an entire batch took approximately 30 h, which was significantly shorter than the traditional batch process. During a single batch, the columns in the Protein A chromatography process were cycled more than 20 times. Longer batch processing times combined with shorter process cycle times could significantly increase the number of column cycles in a batch, thereby promoting disposable bioprocessing.

Continuous manufacturing in combination with a perfusion bioreactor process has also been explored, albeit not in a fully disposable format [5]. Typically, perfusion processes are operated for 20–60 days (160–480 h). Considering the lower titer that comes from perfusion processes, the process cycle times in the chromatography process by nature will be longer and hence, it is realistic to design the process such that the columns reach their maximum lifetime within a single perfusion campaign.

A fully disposable implementation of a multicolumn chromatography technology should also consider the format in which the chromatographic media is provided. Prepacked columns offer significant advantages. In a continuous multicolumn chromatography process, consistency of the column packing is more important than the absolute HETP and asymmetry, since inconsistency in column packing will translate into column-to-column variations in comparing the elution repetitive peaks. It has, however, been demonstrated that multicolumn chromatographic capture processes are quite robust and can handle significant variations in column packing without observing significant effects in performance [13–15]. One of the main advantages of prepacked columns is the ease of use, since it eliminates column packing operations and associated testing.

In this respect, membrane adsorbers also offer an attractive combination with continuous multicolumn chromatography. The main disadvantage of membrane adsorbers for capture chromatography is the relatively low binding capacity. In continuous processes, the impact of binding mainly affects the buffer consumption. The total volume of chromatographic media that is required to run the process is governed by mass transfer kinetics. In this respect, membrane adsorbers are superior to particulate media. The use of membrane adsorbers in combination with BioSMB[®] Technology for capture processes has been evaluated. These studies confirm the significant gain in specific productivity as well as the

impact on buffer consumption [1,16]. The use of membrane adsorbers in combination with a multicolumn chromatography system leads to very short cycle times. This facilitates designing the process in a completely single-use format in a single batch, even in combination with fed batch cultivation processes at moderate expression levels.

2.6

Case Studies

In order to evaluate the impact of continuous bioprocessing and the potential of transforming such process into a fully disposable format, a process design for a 2000 l fed batch bioreactor that produces 4 mg ml^{-1} monoclonal antibody has been considered. With an overall yield of 85%, this would allow a total manufacturing capacity of 136 kg purified drug substance per year in 20 batches (Table 2.4).

Most steps for such process scale are available in disposable format already. The comparison therefore is limited to the two chromatography steps (Protein A and cation exchange). In this comparison, the dynamic binding capacity for the Protein A chromatography media is set at 35 mg ml^{-1} for the batch process. The continuous process was scheduled to run for 12 h on all unit operations. This would result in a total batch processing time of approximately 15 h (assuming approximately 40–45 min average delay time in starting up subsequent unit operations).

A similar process design can be made for a perfusion bioreactor. For a 1000 l bioreactor that has a perfusion rate of 1 bioreactor volume/day and a (steady state) output of 0.85 mg l^{-1} . This compares to an annual manufacturing capacity of approximately 130 kg, assuming six perfusion campaigns per year (30 days each). The results in Table 2.5 are based on the assumption that the downstream processing cascade truly runs fully continuously.

Table 2.4 Key figures for the two chromatography steps in a fed batch cultivation process with a 2000 l bioreactor at 4 mg ml^{-1} expression level.

	Batch	Continuous
Protein A chromatography		
• Cycles per batch	4	15
• Column size	60 cm ID × 20 cm H	14 cm ID × 8 cm H (8 columns)
• Resin volume	57 l	10.2 l (total)
Cation exchange		
• Cycles per batch	4	23
• Column size	40 cm ID × 20 cm H	10 cm ID × 8 cm H (8 columns)
• Resin volume	25 l	4.9 l (total)

Table 2.5 Key figures for the two chromatography steps in a perfusion process with a 1000 l bioreactor at 1.25 mg ml⁻¹ expression level.

	Batch	Continuous
Protein A chromatography		
• Cycles per batch	2 per day	8 per day
• Column size	30 cm ID × 20 cm H	10 cm ID × 5.6 cm H (6 columns)
• Resin volume	14 l	2.54 l (total)
Cation exchange		
• Cycles per batch	2 per day	35 per day
• Column size	20 cm ID × 20 cm H	2.5 cm ID × 7 cm H (8 columns)
• Resin volume	6.3 l	315 ml (total)

In both scenarios, the chromatographic media goes through many more cycles in the continuous process than in the batch process. As a result, disposability of the entire fluid path, including columns, becomes an economically viable option much sooner than in the batch process. For the perfusion bioreactor, one should actually consider installing larger columns than shown in Table 2.5 in order to slow the process down and allow for fewer cycles per day. In this way, the process can be designed to meet the maximum lifetime of the columns at the end of the perfusion campaign. If we assume a 30 days perfusion campaign, the Protein A columns would have to be 20% larger and the cation exchange columns would have to be approximately five times larger than shown in Table 2.5 in order to reach 200 cycles during one campaign. Even with this effect taken into account, the continuous process requires significantly less chromatographic media than the batch downstream process.

These two case studies focus only on the impact of continuous processing on the chromatography steps in order to assess the viability of transforming these steps into a disposable process. The impact on other steps is not addressed in detail, but it is clear that the system volume of all other continuous unit operations will also be significantly smaller compared to their batch equivalents.

2.7

Regulatory Aspects

The US FDA has repeatedly confirmed that continuous processes should be a viable option for biopharmaceutical manufacturing (e.g., Ref. [17]). The advantages of continuous processing in terms of process control should also translate into improved product quality control. In spite of the encouraging messages from the authorities, there are a few concerns that need to be taken into account in the development and filing of products manufactured in continuous processes.

First of all, the definition of a lot and/or batch needs to be considered. Regulators have repeatedly stated that the cGMP guidance does not give any restriction that would suggest that the definition of a lot or a batch is not compatible with continuous processing. As a result, any manufacturer can choose to link the definition of a lot or batch to a certain amount of product produced, a certain time frame or to the use of raw materials or consumables. It makes sense to perform a risk analysis for the sizes of the lot and batch as to balance the costs associated with release testing against the chances and the impact of failure and recalls.

Among the most common reasons for batch rejections, one can find contamination, operator error and equipment error [18]. In continuous processing, the systems by nature will be fully automated. This will lower the chance of operator error. Over the past five years, the average number of batch failures decreased by almost 50% [19]. This was mainly attributed to improved process development and the adoption of the principles of Quality by Design and Process Analytical Technologies. These concepts are natural for continuous processing.

The chance of contamination in a continuous process may be lower due to the fact that there will be fewer and shorter hold steps in which the process solution is held stagnant for a certain period. In general, most process solutions are flowing most of the time and hence the chance of a contamination to occur is smaller. This is further supported by the fact that continuous equipment by nature will have a superior segregation of process solutions. This is an automatic consequence of the requirement of continuous flow of the product solution. For chromatography equipment, this is schematically shown in Figure 2.6. This

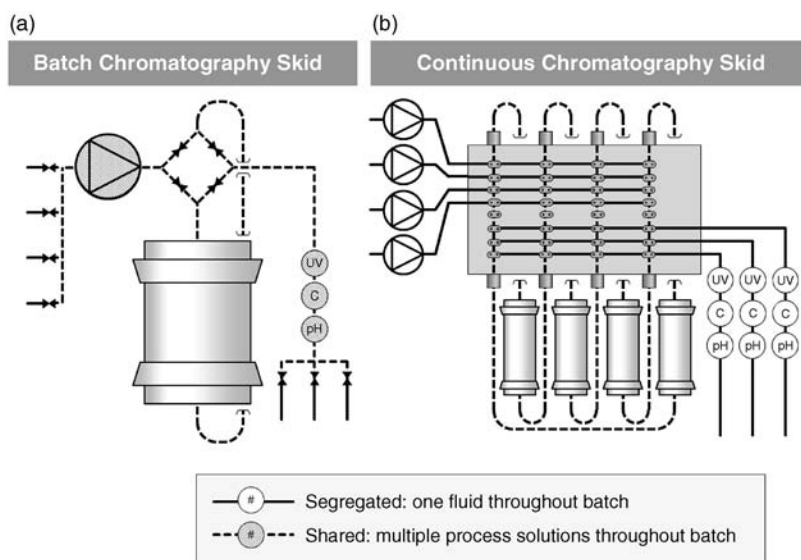


Figure 2.6 Segregation of process solutions in (a) a batch chromatography system and (b) a continuous chromatography system.

Table 2.6 Summary of segregated and shared flow paths in the batch and continuous systems.

Process step	Batch	Continuous
Buffer selection valves	Shared	NA
Pump(s)	Shared	Segregated
Sensors (inlet)	Shared	Segregated
Column bypass valves	Shared	NA
Integrated valve system	NA	Partially shared
Sensors (outlet)	Shared	Segregated
Outlet selection valves	Shared	NA

figure displays a schematic overview of the flow path in a batch chromatography process and in the BioSMB® system. The shared flow paths are shown as dotted lines, whereas all flow paths that are uniquely assigned to a single process solution throughout the entire process cycle are shown as solid lines. The diagrams clearly show how channels and instruments (pumps and sensors) are in contact with all process solutions in the batch process, whereas most channels and all instruments are exclusively in contact with a single process solution in the continuous process. In the continuous system, only the channels to and from the columns – and of course the columns themselves – are in contact with all process solutions, whereas the rest of the entire flow path is dedicated to a single process solution.

The shared and segregated flow paths in the batch and continuous chromatography systems as shown in Figure 2.6 are summarized in Table 2.6.

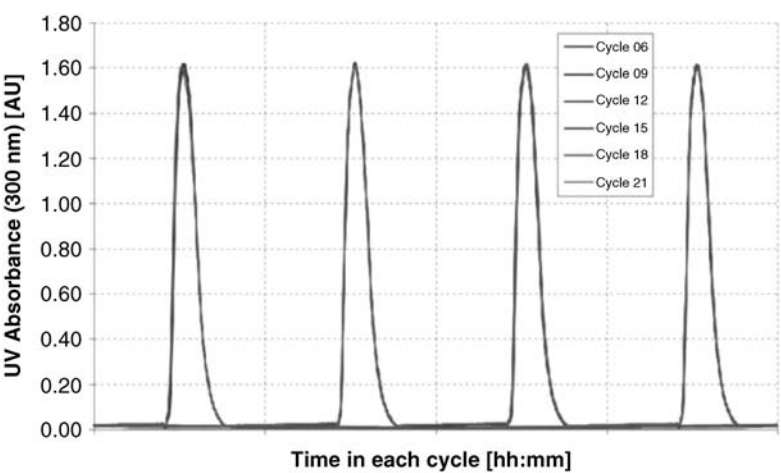


Figure 2.7 Repetitive elution peaks for a four-column chromatography process, showing the elution peaks for six selected cycles overlaid.

Another advantage that continuous processing has to offer is the vast amount of data that is being recorded during the run. In traditional chromatography, the Protein A capture step may only be operated two or three times during a batch. In continuous chromatography, the Protein A capture step can involve hundreds of elution peaks. These repetitive signals provide a very valuable source of information to assess whether or not the system is operating according to its desired (steady) state. An example of such repetitive elution cycles is shown in Figure 2.7 for a four-column chromatography process, displaying the elution peaks of six cycles overlaid.

With a continuous flow of such valuable information, any deviation from the desired (steady) state can be immediately detected. This should favor the risk ranking according to the Failure Mode and Effect Analysis (FMEA) principles.

2.8

Conclusions

The field of biopharmaceutical manufacturing is gradually maturing. As a result, certain manufacturing platforms will gradually evolve into more efficient approaches and as part of this evolution, continuous production is most likely to become a reality in the next decade. Most of the necessary building blocks to implement an integrated continuous biomanufacturing platform are already established and the missing elements are currently being developed. The most important technology to support integrated continuous biomanufacturing is continuous multicolumn chromatography.

One of the key advantages of continuous processing is that the size of most bioprocess systems will be significantly smaller than their batch equivalents. As a result, a disposable format becomes a viable option, even for those steps for which disposable technologies were less attractive in the batch processing platform. Combined with the rapid cycling of media, which is an inherent feature of multicolumn chromatography, the entire fluid path including the columns can be transformed into a viable disposable solution. On the other hand, disposable technologies can mitigate the concerns related to the higher complexity of continuous bioprocessing equipment. The BioSMB® is an example of a technology that can bring this promise.

There is good reason to assume that there will also continue to be batch and hybrid processes. A good example of a viable hybrid process is the combination of a fed batch bioreactor in combination with a (partly) continuous downstream process. The most attractive combination between batch and continuous processing systems depends on an analysis of the risks and benefits of such combination.

From regulatory point of view, there is no objection to continuous processing. Based on a preliminary technical analysis of continuous manufacturing technologies, there seems to be little or no reason to believe that the hurdles will be significantly larger than for batch processes. The same principles of risk-based

validation apply and in certain areas the risks associated with continuous biomanufacturing technologies may actually be lower than the batch equivalents.

Many companies in the biopharmaceutical industries have launched lean six sigma initiatives to streamline their operations. The most notable textbook in the field of Lean Six Sigma is “The Toyota Way” [20]. This book, and the principles of lean manufacturing as implemented at the Toyota car company, has been the foundation for the modern Lean Six Sigma philosophies. With this in mind, one would expect that the adoption of continuous flow manufacturing as practiced by modern car manufacturing would be a logical next step in the biopharmaceutical industry.

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